

ENGR 689 - Multimodal LLM Agents (Fall 2026): Quiz 3

Don't forget to write your name on the paper you submit!

Each question has only one correct option.

10 questions Duration: 25 Minutes

- Q1 (1%)** In a Vision Transformer (ViT), what is the main purpose of splitting an image into patches?
- A. To generate captions for each image region independently.
 - B. To align every patch directly with a word in the caption.
 - C. To convert image regions into a sequence of tokens for a Transformer.
 - D. To reduce every image to a single scalar feature.
- Q2 (1%)** A ViT uses a fixed patch size of 14×14 . The input resolution is increased from 336×336 to 672×672 . Assuming the computational cost of self-attention scales quadratically with the number of visual tokens, approximately how does the self-attention cost change?
- A. It increases by $2\times$.
 - B. It increases by $4\times$.
 - C. It increases by $8\times$.
 - D. It increases by $16\times$.
- Q3 (1%)** What is the main training objective used by CLIP to learn aligned vision-language representations?
- A. Predicting the next word given an image.
 - B. Reconstructing masked image patches from text.
 - C. Predicting bounding boxes from natural-language queries.
 - D. Increasing similarity of matched image-text pairs while decreasing similarity of mismatched pairs.
- Q4 (1%)** Suppose a pretrained vision encoder produces visual features that cannot be directly consumed by a pretrained LLM. What is the primary role of a multimodal connector, such as the projection layer in LLaVA?
- A. To classify images before sending only the class label to the language model.
 - B. To transform and bridge visual features into representations that can be used by the language model.
 - C. To replace the pretrained vision encoder with a language model.
 - D. To generate additional image-text training pairs automatically.
- Q5 (1%)** A model correctly answers that an image contains a red cup, but cannot identify which image region contains the cup. Which capability is most directly missing?
- A. Visual grounding.
 - B. Language modeling.
 - C. Contrastive pretraining.
 - D. Image captioning.

Q6 (1%) What does visual instruction tuning mainly teach a multimodal LLM to do?

- A. Learn visual features without using any language supervision.
- B. Train only the image decoder for higher-resolution generation.
- C. Convert all images into fixed category labels.
- D. Follow natural-language instructions that depend on visual input.

Q7 (1%) In a text-to-image diffusion model, what is the main role of the text representation during image generation?

- A. To directly replace the noisy image representation with the final image.
- B. To condition the denoising process so that the generated image is consistent with the text prompt.
- C. To classify the generated image into one of several fixed categories after generation.
- D. To determine the number of denoising steps solely from the length of the text prompt.

Q8 (1%) Consider two systems that can both analyze an input image and generate a new image. System A uses a vision-language model for understanding and a separate image-generation model, with a controller passing information between them. System B is trained as a unified multimodal model that supports both understanding and generation. Which statement best distinguishes System B from System A?

- A. System B can condition image generation on information extracted from an input image, whereas System A cannot.
- B. System B can support multi-turn interactions involving both images and text, whereas System A cannot.
- C. System B learns understanding and generation within shared model parameters and representations, rather than relying on explicit routing between separately trained models.
- D. System B can produce both textual and visual outputs, whereas System A can produce only one output modality.

Q9 (1%) An agent is building a database that matches runner faces with bib numbers. The face and bib detectors work correctly, but the vertical-distance rule often matches a bib to the wrong runner when people overlap. Which next step best reflects an agentic workflow?

- A. Store all current matches because both detectors completed successfully.
- B. Process more video frames using the same matching rule without evaluating its errors.
- C. Inspect the intermediate detections, revise the matching method using additional spatial or tracking information, and test it again.
- D. Replace the full workflow with a single zero-shot prompt asking for the final database.

Q10 (1%) A vision-language model (VLM)-based robot receives the instruction: “Use the broom to move the trash to the right side of the table.” Which option correctly describes the roles of high-level and low-level reasoning?

- A. High-level reasoning selects exact grasp and motion keypoints, while low-level reasoning determines the sequence of semantic subtasks.
- B. High-level reasoning decomposes the instruction into semantic subtasks, while low-level reasoning grounds each subtask into grasp points, contact points, and motion waypoints.
- C. High-level reasoning generates only segmentation masks, while low-level reasoning interprets the complete user instruction without using the subtasks.
- D. High-level and low-level reasoning independently generate the same robot motion, and the robot selects one through majority voting.